

Mariam Ahmed El-Sebaay

Mechatronics Engineering Student | Embedded Systems & Robotics Enthusiast

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🌐 Mariam El-Sebaay 🐙 github.com/mariaam2005 🔗 mariaam2005.github.io/portfolio

Objective

Motivated Mechatronics Engineering student with a strong interest in Embedded Systems, Robotics, and Automation. Seeking opportunities to apply technical and problem-solving skills through hands-on projects, practical training, and real-world engineering challenges.

Education

Bachelor of Mechatronics Engineering, Egyptian Chinese University (ECU) 09/2023 – Present
Level 3 Mechatronics Engineering Student | Current GPA: 3.8 / 4.0

Training & Activities

Electrical & Avionics Aircraft Systems Training, EgyptAir 08/2026 – 09/2026

- Studied aircraft systems including communication, navigation, autoflight, electrical, hydraulic, fuel, flight control, landing gear, APU, and fire & smoke systems.
- Gained hands-on experience in aircraft hangars with aircraft units, engines, and equipment.

Electrical & Mechanical Training, CNTC.egypt 07/2026 – 08/2026

- Studied power station operation, turbine types, structures, and operating principles.
- Visited North Cairo and Shoubra Power Stations to observe electrical and mechanical systems.

Technical Field Visit, Cairo Metro Control Center 08/2025 – 08/2025

- Observed maintenance of electrical and mechanical systems.
- Learned about control systems, power, and safety procedures.

Projects

Autonomous Medical Delivery Robot

- Developing an autonomous medical delivery robot integrated with a Flutter mobile application and Firebase Realtime Database for real-time hospital delivery management.
- Implementing RFID authentication, ESP32 communication, and AI-based automation for secure medicine delivery and smart healthcare assistance.

Charging Station

- Developed a smart hybrid charging station integrating solar energy and grid power for safe and efficient battery charging.
- Implemented ESP32-based real-time monitoring and control system with voltage, current, and temperature sensors for multi-slot battery management.

Delta Robot with Computer Vision – Pick & Place Automation System

- Developed a Delta robot for automated pick-and-place operations.
- Integrated computer vision for object detection and sorting.

- Applied motion control and kinematic design concepts.

Smart Home Automation System

- Designed an automated system to control lighting and fan speed based on temperature.
- Integrated LM35, LDR, and flame sensor with buzzer alerts.
- Used ATmega32 with ADC, PWM, and GPIO drivers.

Vehicle Fault Detection System

- Designed dual ECU system communicating via UART.
- Monitored faults using sensors, EEPROM, and LCD.

Courses

ROS & Robotics Diploma, *Edges for Training* 07/2026 – Present

Linux Administration Course, *Edges for Training* 06/2026 – 07/2026

- Covered Linux fundamentals, CLI, users & permissions, Bash scripting, process management, and networking.
- Completed hands-on labs , practical exercises and a final project.

Embedded Systems Diploma, *Edges for Training* 06/2025 – 09/2025

- Studied Embedded C, AVR interfacing, RTOS basics.
- Completed multiple hardware-based projects.

Embedded Systems Workshop, *Egyptian Chinese University* 06/2025 – 07/2025

- Gained hands-on experience with microcontrollers and basic interfacing.

Mechanical Design Workshop, *Pascal Academy* 01/2025 – 02/2025

- Learned basics of mechanical design and engineering drawing.

Technical Skills

Embedded Systems

Arduino, AVR, ATmega32, Embedded C, GPIO, UART, RTOS

Mobile & Software Development

Flutter, Qt, Firebase, Git/GitHub, Linux

Mechanical Design

SolidWorks, Engineering Drawing

Programming Languages

C, C++, Python, Dart

Database Management

SQL,SQLite,Cloud Firestore

Soft Skills

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|----------------------|-------------------------------|
| • Project Management | • Team Leadership |
| • Problem Solving | • Time Management |
| • Teamwork | • Communication |
| • Adaptability | • Bilingual: Arabic & English |